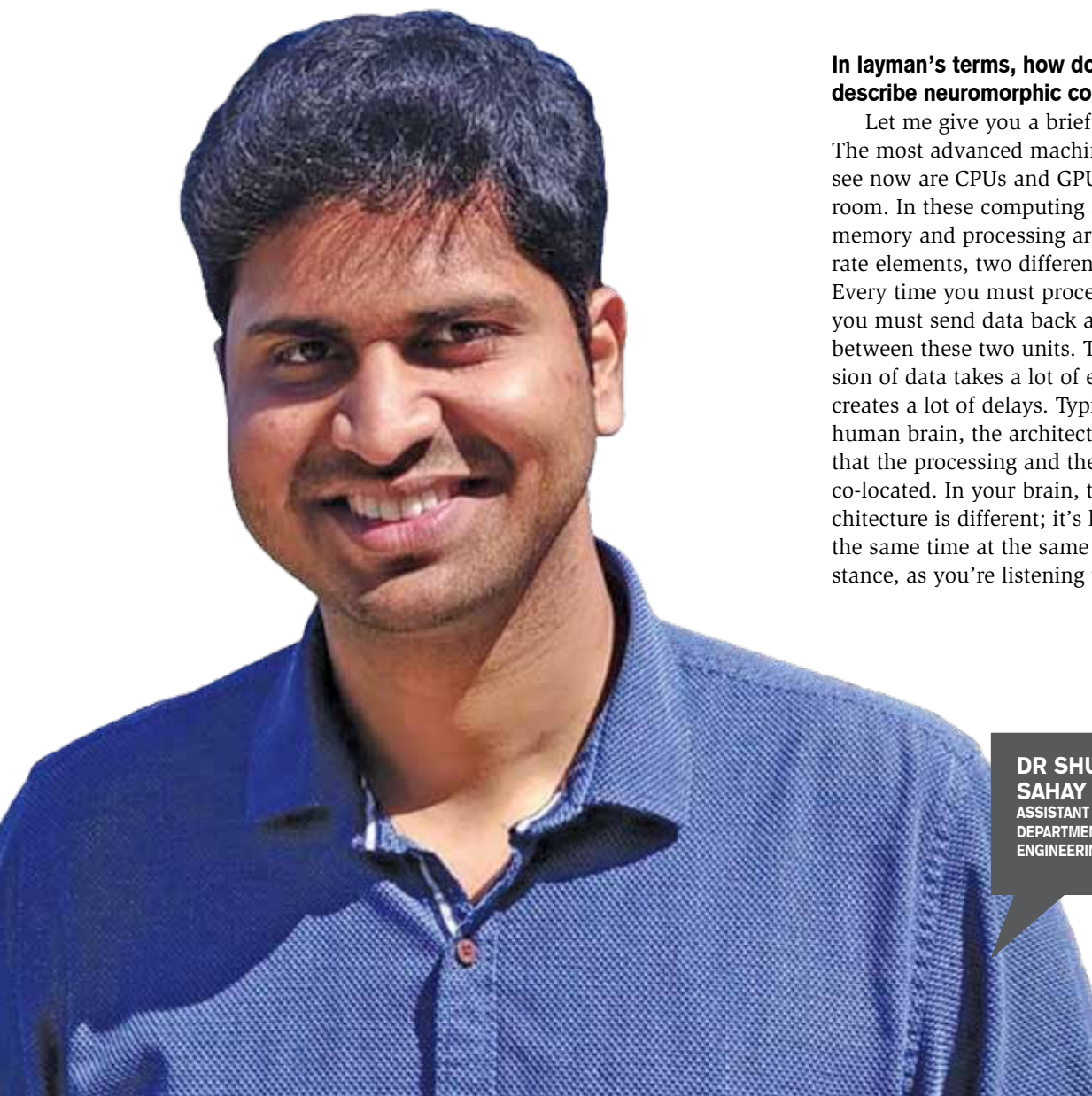


NEUROMORPHIC COMPUTING: It's Not Sci-Fi Anymore

What is Neuromorphic Computing? What are its applications? And, what's happening in India on this front? We were fortunate to get connected with Dr Shubham Sahay, Assistant Professor at IIT Kanpur, who specialises in neuromorphic computing and hardware security. Here are some extracts of his interaction with Aaryaa Padhyegurjar from EFY



In layman's terms, how do you describe neuromorphic computing?

Let me give you a brief example. The most advanced machines that you see now are CPUs and GPUs in a server room. In these computing systems, memory and processing are two separate elements, two different entities. Every time you must process any data, you must send data back and forth between these two units. That transmission of data takes a lot of energy and creates a lot of delays. Typically, in your human brain, the architecture is such that the processing and the memory are co-located. In your brain, the whole architecture is different; it's happening at the same time at the same place. For instance, as you're listening to a conver-

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